

# Michael Soprano, PhD

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🐙 <https://github.com/Miccighel>



## Education

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| Nov. 2019 – Jan. 2023 | <b>PhD in Computer Science, Mathematics and Physics</b> , Cum Laude. 35 <sup>th</sup> cycle, University of Udine, Italy.                                |
| Mar. 2015 – Mar. 2018 | <b>Master Degree in Computer Science</b> , 109/110, University of Udine, Italy.                                                                         |
| Oct. 2011 – Mar. 2015 | <b>Bachelor Degree in Web Technologies and Multimedia</b> , 101/110, University of Udine, Italy.                                                        |
| Sep. 2006 – Jun. 2011 | <b>High School Diploma</b> Head Technician Industrial Expert – Computer Science Specialization, 77/100, I.S.I.S. R. D'Aronco, Gemona del Friuli, Italy. |

## Work Experience

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| Jun. 2025 – Present   | <b>Postdoctoral Researcher.</b> Department of Mathematics, Computer Science and Physics (DMIF), University of Udine, Italy.<br>Research contract pursuant to Article 22 of Law 240/2010, regulated by the new University regulation approved on April 30 2025. Research on information and disinformation, with focus on the automatic evaluation of information truthfulness, experimentation with detection models, and definition of evaluation metrics.                                |
| Mar. 2025 – May 2025  | <b>Self-employed Worker.</b> Department of Economic and Statistical Sciences (DIES), University of Udine, Italy.<br>Design, implementation, and evaluation of a conversational system for accessing regional public healthcare services; prototype development, experimental evaluation, and data analysis within the inter-departmental project “Governance and Digital Public Administration.”                                                                                           |
| Dec. 2023 – May 2025  | <b>Postdoctoral Researcher.</b> DMIF, University of Udine.<br>PRIN 2022 project – “The Measure of Truth: An Evaluation-Centered Machine-Human Hybrid Framework for Assessing Information Truthfulness” (Code 20227F2ZN3, CUP G53D23002800006), funded by the EU – Next Generation EU – PNRR M4 C2 I1.1. Definition of an evaluation framework for misinformation-detection systems: creation of labelled datasets, bias prevention, and metrics that account for information truthfulness. |
| Mar. 2023 – Nov. 2023 | <b>Postdoctoral Researcher.</b> DIES, University of Udine.<br>Research on eGovernment applications based on conversational agents for the Italian public administration, with particular emphasis on the healthcare domain.                                                                                                                                                                                                                                                                |
| Jan. 2022 – Mar. 2022 | <b>Self-employed Worker.</b> Polytechnic Department of Engineering and Architecture (DPIA), University of Udine.<br>Development of a web application for the comparability and reproducibility of research results in combinatorial optimization.                                                                                                                                                                                                                                          |
| Jun. 2018 – Oct. 2019 | <b>Research Fellow.</b> DMIF, University of Udine.<br>Research project co-funded by SISSA Medialab: design, implementation, and evaluation of an ecosystem providing an alternative to traditional peer review in scholarly publishing.                                                                                                                                                                                                                                                    |

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| Jun. 2017 – Oct. 2019 | <b>Self-employed Worker.</b> Datamantix S.r.l., Udine, Italy.<br>Design, development, and maintenance of various web applications and a chat-bot as a full-stack developer. |
| Aug. 2017 – Feb. 2018 | <b>Self-employed Worker.</b> Archeido S.r.l., Udine, Italy.<br>Development of a web application for managing and visualizing financial assets as a full-stack developer.    |
| Nov. 2015 – Feb. 2016 | <b>Self-employed Worker.</b> Troublezine.<br>Design, development, and maintenance of a music news and events blog as a full-stack web developer.                            |

## Visits

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| Mar. 2022 – Jul. 2022 | <b>Visiting Student</b> , Centrum Wiskunde & Informatica (CWI), Human-Centered Data Analytics Group, Amsterdam, The Netherlands. Research visit during the PhD programme. |
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## Presentations

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| Jun. 2026 | <a href="#">WORKSHOP TALK.</a> <b>Crowdsourced Fact-Checking for Misinformation: A Journey Through Nine Years of Research (2018–2026).</b> <i>Workshop “AI and Disinformation”, held as part of the 6th National CINI Conference on Artificial Intelligence (Ital-IA 2026), June 18, 2026. Rome, Italy.</i><br><a href="#">Slides</a>                                                     |
| Jun. 2026 | <a href="#">POSTER.</a> <b>Crowdsourced Keypoint Matching for Region of Interest Identification in Cross-Stain Whole Slide Image Registration.</b> <i>22nd European Congress on Digital Pathology (ECDP 2026), June 18, 2026. Graz, Austria.</i><br><a href="#">Poster PDF</a>                                                                                                            |
| Apr. 2026 | <a href="#">WORKSHOP TALK.</a> <b>Beyond Seeing Is Believing: On Crowdsourced Detection of Audiovisual Deepfakes.</b> <i>The 6th Workshop on Reducing Online Misinformation through Credible Information Retrieval (ROMCIR 2026), held as part of The 48th European Conference on Information Retrieval (ECIR 2026), April 2, 2026. Delft, The Netherlands.</i><br><a href="#">Slides</a> |
| Mar. 2026 | <a href="#">POSTER.</a> <b>Analyzing AI Evaluation Benchmarks Through Information Retrieval and Network Science.</b> <i>The 48th European Conference on Information Retrieval (ECIR 2026), March 30, 2026. Delft, The Netherlands.</i><br><a href="#">Poster PDF</a>                                                                                                                      |
| Mar. 2026 | <a href="#">POSTER.</a> <b>Large Language Models as Assessors: On the Impact of Relevance Scales.</b> <i>The 48th European Conference on Information Retrieval (ECIR 2026), March 30, 2026. Delft, The Netherlands.</i><br><a href="#">Poster PDF</a>                                                                                                                                     |
| Jul. 2025 | <a href="#">POSTER.</a> <b>AIDME: A Scalable, Interpretable Framework for AI-Aided Scoping Reviews.</b> <i>The 15th International Conference on Innovative Concepts and Theories in Information Retrieval, July 18, 2025. Padua, Italy.</i><br><a href="#">Poster PDF</a>                                                                                                                 |
| Jul. 2025 | <a href="#">CONFERENCE TALK.</a> <b>The Magnitude of Truth: On Using Magnitude Estimation for Truthfulness Assessment.</b> <i>The 48th International ACM SIGIR Conference on Research and Development in Information Retrieval, July 16, 2025. Padua, Italy.</i><br><a href="#">Slides</a>                                                                                                |
| Jun. 2025 | <a href="#">WORKSHOP TALK.</a> <b>AIDME: A Scalable, Interpretable Framework for AI-Aided Scoping Reviews.</b> <i>Workshop “AI, Misinformation and Disinformation”, held as part of the 5th National CINI Conference on Artificial Intelligence (Ital-IA 2025), June 23, 2025. Trieste, Italy.</i><br><a href="#">Slides</a>                                                              |

- Feb. 2025 **CONFERENCE TALK. Evaluation of Crowdsourced Peer Review using Synthetic Data and Simulations.** *21st Conference on Information and Research Science Connecting to Digital and Library Science (IRCDL 2025)*, February 20, 2025. Udine, Italy.  
[Slides](#)
- Sep. 2024 **DEMO PRESENTATION. Deploying Complex Crowdsourcing Tasks Without Fear: An Introduction to Crowd\_Frame.** *The 3rd Italian Conference on Big Data and Data Science (ITADATA 2024)*, September 17, 2024. Pisa, Italy.  
[Slides](#)
- Oct. 2023 **WORKSHOP TALK. Towards a Conversational-Based Agent for Healthcare.** *CARE: Conversational Agents and Responsible Ethics in Healthcare Workshop, held as part of the European Federation of Medical Informatics Special Topic Conference 2023 (EFMI STC 2023)*, October 26, 2023. Turin, Italy.  
[Slides](#)
- May 2023 **WORKSHOP TALK. Towards a Conversational-Based Agent for Health Services.** *Workshop "AI for Public Administration", held as part of the 3rd National CINI Conference on Artificial Intelligence (Ital-IA 2023)*, May 29, 2023. Pisa, Italy.  
[Slides](#)
- Feb. 2022 **CONFERENCE TALK. Crowd\_Frame: A Simple and Complete Framework to Deploy Complex Crowdsourcing Tasks Off-the-Shelf.** *The 15th ACM International Conference on Web Search and Data Mining (WSDM 2022)*, February 23, 2022. Phoenix, U.S.A. (remote).  
Held remotely as a pre-recorded contribution due to the COVID-19 pandemic.  
[Slides](#) [Video](#)
- Oct. 2020 **CONFERENCE TALK. The COVID-19 Infodemic: Can the Crowd Judge Recent Misinformation Objectively?** *The 29th ACM International Conference on Information and Knowledge Management (CIKM 2020)*, October 19, 2020. Galway, Ireland (remote).  
Held remotely as a pre-recorded contribution due to the COVID-19 pandemic.  
[Slides](#) [Video](#)
- Jul. 2020 **CONFERENCE TALK. Can The Crowd Identify Misinformation Objectively? The Effects of Judgment Scale and Assessor's Background.** *The 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR 2020)*, July 25, 2020. Xi'an, China (remote).  
Held remotely as a pre-recorded contribution due to the COVID-19 pandemic.  
[Slides](#) [Video](#)
- Sep. 2019 **WORKSHOP TALK. Bias and Fairness in Effectiveness Evaluation by Means of Network Analysis and Mixture Model.** *Italian Information Retrieval Workshop (IIR 2019)*, September 16, 2019. Padua, Italy.  
[Slides](#)

## Publications

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### Journal Articles

- 1 Francesca Da Ros, **Michael Soprano**, Luca Di Gaspero e Kevin Roitero. «Large Language Models for Combinatorial Optimization: A Systematic Review». In: *ACM Computing Surveys* 58.11 (ago. 2026). Journal Ranks: Journal Citation Reports (JCR) Q1 (2024), Scimago (SJR) Q1 (2024). ISSN: 0360-0300. DOI: 10.1145/3801961.
- 2 Eddy Maddalena, Stefano Mizzaro, Kevin Roitero, Marco Viviani, David La Barbera, Sandip Modha, Gabriella Pasi, Francesca Da Ros e **Michael Soprano**. «Report on the 14th Italian Information Retrieval Workshop (IIR 2024)». In: *SIGIR Forum* 58.2 (dic. 2024), pp. 1–13. ISSN: 0163-5840. DOI: 10.1145/3722449.3722464.
- 3 Damiano Spina, Kevin Roitero, Stefano Mizzaro, Vincenzo Della Mea, Francesca Da Ros, **Michael Soprano**, Hafsa Akebli, Alex Falcon, Mehdi Fasihi, Alessio Fiorin, David La Barbera, Daniele Lizzio Bosco, Riccardo Lunardi, Alberto Marturano, Zaka-Ud-Din Muhammad, Francesco Nascimben, Moritz Nottebaum, Massimiliano Pascoli, Mihai Horia Popescu, Laura Rasotto, Mubashara Rehman, Francesco Taverna, Biagio Tomasetig e Alessandro Tremamunno. «Report on the Hands-On PhD Course on Responsible AI from the Lens of an Information Access Researcher». In: *SIGIR Forum* 58.2 (dic. 2024), pp. 1–61. ISSN: 0163-5840. DOI: 10.1145/3722449.3722456.

- 4 **Michael Soprano**, Kevin Roitero, Ujwal Gadiraju, Eddy Maddalena e Gianluca Demartini. «Longitudinal Loyalty: Understanding The Barriers To Running Longitudinal Studies On Crowdsourcing Platforms». In: *ACM Transactions on Social Computing* 7.1–4 (set. 2024). Journal Rank: Scimago (SJ) Q2 (2023). DOI: 10.1145/3674884.
- 5 David La Barbera, Eddy Maddalena, **Michael Soprano**, Kevin Roitero, Gianluca Demartini, Davide Ceolin, Damiano Spina e Stefano Mizzaro. «Crowdsourced Fact-checking: Does It Actually Work?» In: *Information Processing & Management* 61.5 (lug. 2024). Journal Ranks: Journal Citation Reports (JCR) Q1 (2023), Scimago (SJ) Q1 (2023), p. 103792. ISSN: 0306-4573. DOI: 10.1016/j.ipm.2024.103792.
- 6 **Michael Soprano**, Kevin Roitero, David La Barbera, Davide Ceolin, Damiano Spina, Gianluca Demartini e Stefano Mizzaro. «Cognitive Biases in Fact-Checking and Their Countermeasures: A Review». In: *Information Processing & Management* 61.3 (2024). Journal Ranks: Journal Citation Reports (JCR) Q1 (2022), Scimago (SJ) Q1 (2022), p. 103672. ISSN: 0306-4573. DOI: 10.1016/j.ipm.2024.103672.
- 7 Kevin Roitero, David La Barbera, **Michael Soprano**, Gianluca Demartini, Stefano Mizzaro e Tetsuya Sakai. «How Many Crowd Workers Do I Need? On Statistical Power When Crowdsourcing Relevance Judgments». In: *ACM Transactions on Information Systems* (mag. 2023). Journal Ranks: Journal Citation Reports (JCR) Q1 (2021), Scimago (SJ) Q1 (2021). ISSN: 1046-8188. DOI: 10.1145/3597201.
- 8 Erik Brand, Kevin Roitero, **Michael Soprano**, Afshin Rahimi e Gianluca Demartini. «A Neural Model to Jointly Predict and Explain Truthfulness of Statements». In: *Journal of Data and Information Quality* 15.1 (dic. 2022). Journal Ranks: Journal Citation Reports (JCR) Q3 (2021), Scimago (SJ) Q2 (2021). ISSN: 1936-1955. DOI: 10.1145/3546917.
- 9 Davide Ceolin, Giuseppe Primiero, **Michael Soprano** e Jan Wielemaker. «Transparent Assessment of Information Quality of Online Reviews Using Formal Argumentation Theory». In: *Information Systems* 110 (lug. 2022). Journal Ranks: Journal Citation Reports (JCR) Q2 (2021), Scimago (SJ) Q1 (2021), p. 102107. ISSN: 0306-4379. DOI: 10.1016/j.is.2022.102107.
- 10 **Michael Soprano**, Kevin Roitero, David La Barbera, Davide Ceolin, Damiano Spina, Stefano Mizzaro e Gianluca Demartini. «The Many Dimensions of Truthfulness: Crowdsourcing Misinformation Assessments on a Multidimensional Scale». In: *Information Processing & Management* 58.6 (nov. 2021). Journal Ranks: Journal Citation Reports (JCR) Q1 (2021), Scimago (SJ) Q1 (2021), p. 102710. ISSN: 0306-4573. DOI: 10.1016/j.ipm.2021.102710.
- 11 Kevin Roitero, **Michael Soprano**, Beatrice Portelli, Massimiliano De Luise, Damiano Spina, Vincenzo Della Mea, Giuseppe Serra, Stefano Mizzaro e Gianluca Demartini. «Can the Crowd Judge Truthfulness? A Longitudinal Study on Recent Misinformation About COVID-19». In: *Personal and Ubiquitous Computing* 27.1 (feb. 2021). Journal Ranks: Journal Citation Reports (JCR) Q2 (2020), Scimago (SJ) Q1 (2021), pp. 59–89. ISSN: 1617-4917. DOI: 10.1007/s00779-021-01604-6.
- 12 Kevin Roitero, **Michael Soprano**, Andrea Brunello e Stefano Mizzaro. «Reproduce and Improve: An Evolutionary Approach to Select a Few Good Topics for Information Retrieval Evaluation». In: *Journal of Data and Information Quality* 10.3 (set. 2018). Journal Rank: Scimago (SJ) Q2 (2018). ISSN: 1936-1955. DOI: 10.1145/3239573.

### Conference and Workshop Papers

- 13 Gaia Simeoni, **Michael Soprano**, Riccardo Lunardi, Kevin Roitero e Stefano Mizzaro. «Analyzing AI Evaluation Benchmarks Through Information Retrieval and Network Science». In: *Advances in Information Retrieval: 48th European Conference on Information Retrieval (ECIR 2026)*. Vol. 16484. Lecture Notes in Computer Science. Conference Rank: CORE A. Cham: Springer Nature Switzerland, mar. 2026, pp. 349–359. ISBN: 978-3-032-21300-6. DOI: 10.1007/978-3-032-21300-6\_25.
- 14 Riccardo Zamolo, Riccardo Lunardi, **Michael Soprano**, Gianluca Demartini, Stefano Mizzaro e Kevin Roitero. «Large Language Models as Assessors: On the Impact of Relevance Scales». In: *Advances in Information Retrieval: 48th European Conference on Information Retrieval (ECIR 2026)*. Vol. 16484. Lecture Notes in Computer Science. Conference Rank: CORE A. Cham: Springer Nature Switzerland, mar. 2026, pp. 338–348. ISBN: 978-3-032-21300-6. DOI: 10.1007/978-3-032-21300-6\_24.
- 15 Riccardo Lunardi, **Michael Soprano**, Paolo Coppola, Vincenzo Della Mea, Stefano Mizzaro e Kevin Roitero. «PILs of Knowledge: A Synthetic Benchmark for Evaluating Question Answering Systems in Healthcare». In: *Proceedings of the 48th International ACM SIGIR Conference on Research and Development in Information Retrieval*. SIGIR '25. Conference Ranks: GGS A++, CORE A\*. Padua, Italy: Association for Computing Machinery, lug. 2025, pp. 3648–3658. ISBN: 9798400715921. DOI: 10.1145/3726302.3730283.

- 16 Kevin Roitero, Dustin Wright, **Michael Soprano**, Isabelle Augenstein e Stefano Mizzaro. «Efficiency and Effectiveness of LLM-Based Summarization of Evidence in Crowdsourced Fact-Checking». In: *Proceedings of the 48th International ACM SIGIR Conference on Research and Development in Information Retrieval*. SIGIR '25. Conference Ranks: GGS A++, CORE A\*. Padua, Italy: Association for Computing Machinery, lug. 2025, pp. 457–467. ISBN: 9798400715921. DOI: 10.1145/3726302.3729960.
- 17 **Michael Soprano**, Sandip Modha, Kevin Roitero, Eddy Maddalena, Marco Viviani, Gabriella Pasi e Stefano Mizzaro. «AIDME: A Scalable, Interpretable Framework for AI-Aided Scoping Reviews». In: *Proceedings of the 2025 International ACM SIGIR Conference on Innovative Concepts and Theories in Information Retrieval (ICTIR)*. ICTIR '25. Padua, Italy: Association for Computing Machinery, lug. 2025, pp. 194–207. ISBN: 9798400718618. DOI: 10.1145/3731120.3744586.
- 18 **Michael Soprano**, Denis Eduard Tapu, David La Barbera, Kevin Roitero e Stefano Mizzaro. «The Magnitude of Truth: On Using Magnitude Estimation for Truthfulness Assessment». In: *Proceedings of the 48th International ACM SIGIR Conference on Research and Development in Information Retrieval*. SIGIR '25. Conference Ranks: GGS A++, CORE A\*. Padua, Italy: Association for Computing Machinery, lug. 2025, pp. 446–456. ISBN: 9798400715921. DOI: 10.1145/3726302.3730091.
- 19 Riccardo Lunardi, Kevin Roitero, **Michael Soprano**, Paolo Coppola, Vincenzo Della Mea e Stefano Mizzaro. «Agent-Based Healthcare Chatbots for Regional System Services: A Case Study in the Friuli-Venezia Giulia Region». In: *Intelligent Health Systems – From Technology to Data and Knowledge*. Vol. 327. Studies in Health Technology and Informatics. Glasgow, Scotland: IOS Press, mag. 2025, pp. 677–681. ISBN: 978-1-64368-485-7. DOI: 10.3233/SHTI250434.
- 20 Francesca Da Ros, Luca Di Gaspero, Marie-Louise Lackner, Nysret Musliu e **Michael Soprano**. «Search Trajectory Networks Applied to a Real-World Parallel Batch Scheduling Problem». In: *Applications of Evolutionary Computation (EvoApplications 2025)*. Vol. 15612. Lecture Notes in Computer Science. Conference Ranks: GGS B, CORE B. Cham: Springer Nature Switzerland, apr. 2025, pp. 68–85. ISBN: 978-3-031-90062-4. DOI: 10.1007/978-3-031-90062-4\_5.
- 21 **Michael Soprano**, Eddy Maddalena, Francesca Da Ros, Maria Elena Zuliani e Stefano Mizzaro. «Evaluation of Crowdsourced Peer Review using Synthetic Data and Simulations». In: *Proceedings of the 21st Conference on Information and Research Science Connecting to Digital and Library Science*. Vol. 3937. CEUR Workshop Proceedings. Udine, Italy: CEUR-WS.org, mar. 2025.
- 22 Kevin Roitero, **Michael Soprano**, David La Barbera, Eddy Maddalena e Stefano Mizzaro. «Enhancing Fact-Checking: From Crowdsourced Validation to Integration with Large Language Models». In: *Proceedings of the 14th Italian Information Retrieval Workshop*. Vol. 3802. CEUR Workshop Proceedings. Udine, Italy: CEUR-WS.org, ott. 2024, pp. 74–77.
- 23 Jaspreet Singh, **Michael Soprano**, Kevin Roitero e Davide Ceolin. «Crowdsourcing Statement Classification to Enhance Information Quality Prediction». In: *Proceedings of the 6th Multidisciplinary International Symposium on Disinformation in Open Online Media (MISDOOM 2024)*. Lecture Notes in Computer Science. Münster, Germany: Springer Nature Switzerland, ago. 2024, pp. 70–85. ISBN: 978-3-031-71210-4. DOI: 10.1007/978-3-031-71210-4\_5.
- 24 **Michael Soprano**, Kevin Roitero, Vincenzo Della Mea e Stefano Mizzaro. «Towards a Conversational-Based Agent for Health Services». In: *Proceedings of the Italia Intelligenza Artificiale - Thematic Workshops co-located with the 3rd CINI National Lab AIIS Conference on Artificial Intelligence*. Vol. 3486. CEUR Workshop Proceedings. Pisa, Italy: CEUR-WS.org, set. 2023, pp. 278–283.
- 25 David La Barbera, **Michael Soprano**, Kevin Roitero, Eddy Maddalena e Stefano Mizzaro. «Fact-Checking at Scale with Crowdsourcing: Experiments and Lessons Learned». In: *Proceedings of the 13th Italian Information Retrieval Workshop*. Vol. 3448. CEUR Workshop Proceedings. Pisa, Italy: CEUR-WS.org, ago. 2023, pp. 85–90.
- 26 Tim Draws, David La Barbera, **Michael Soprano**, Kevin Roitero, Davide Ceolin, Alessandro Checco e Stefano Mizzaro. «The Effects of Crowd Worker Biases in Fact-Checking Tasks». In: *2022 ACM Conference on Fairness, Accountability, and Transparency*. FAccT '22. Seoul, Republic of Korea: Association for Computing Machinery, giu. 2022, pp. 2114–2124. ISBN: 9781450393522. DOI: 10.1145/3531146.3534629.
- 27 **Michael Soprano**, Kevin Roitero, Francesco Bombassei De Bona e Stefano Mizzaro. «Crowd\_Frame: A Simple and Complete Framework to Deploy Complex Crowdsourcing Tasks Off-the-Shelf». In: *Proceedings of the Fifteenth ACM International Conference on Web Search and Data Mining*. WSDM '22. Conference Ranks: GGS A+, CORE A\*. Virtual Event, AZ, USA: Association for Computing Machinery, feb. 2022, pp. 1605–1608. ISBN: 9781450391320. DOI: 10.1145/3488560.3502182.



- 28 Erik Brand, Kevin Roitero, **Michael Soprano** e Gianluca Demartini. «E-BART: Jointly Predicting and Explaining Truthfulness». In: *Proceedings of The 2021 Truth and Trust Online Conference*. Virtual Event, ott. 2021, pp. 18–27.
- 29 Davide Ceolin, Giuseppe Primiero, Jan Wielemaker e **Michael Soprano**. «Assessing the Quality of Online Reviews Using Formal Argumentation Theory». In: *Web Engineering*. Conference Ranks: GGS B-, CORE B. Biarritz, France: Springer International Publishing, mag. 2021, pp. 71–87. ISBN: 978-3-030-74296-6. DOI: 10.1007/978-3-030-74296-6\_6.
- 30 Kevin Roitero, **Michael Soprano**, Beatrice Portelli, Damiano Spina, Vincenzo Della Mea, Giuseppe Serra, Stefano Mizzaro e Gianluca Demartini. «The COVID-19 Infodemic: Can the Crowd Judge Recent Misinformation Objectively?». In: *Proceedings of the 29th ACM International Conference on Information and Knowledge Management*. CIKM '20. Conference Ranks: GGS A+, CORE A. Virtual Event, Ireland: Association for Computing Machinery, ott. 2020, pp. 1305–1314. ISBN: 9781450368599. DOI: 10.1145/3340531.3412048.
- 31 Kevin Roitero, **Michael Soprano**, Shaoyang Fan, Damiano Spina, Stefano Mizzaro e Gianluca Demartini. «Can The Crowd Identify Misinformation Objectively? The Effects of Judgment Scale and Assessor's Background». In: *Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval*. SIGIR '20. Conference Ranks: GGS A++, CORE A\*. Xi'an, China (Virtual): Association for Computing Machinery, lug. 2020, pp. 439–448. ISBN: 9781450380164. DOI: 10.1145/3397271.3401112.
- 32 **Michael Soprano**, Kevin Roitero e Stefano Mizzaro. «Bias and Fairness in Effectiveness Evaluation by Means of Network Analysis and Mixture Models». In: *Proceedings of the 10th Italian Information Retrieval Workshop*. Vol. 2441. CEUR Workshop Proceedings. Padova, Italy: CEUR-WS.org, set. 2019, 6–7.
- 33 **Michael Soprano**, Kevin Roitero e Stefano Mizzaro. «HITS Hits Readersourcing: Validating Peer Review Alternatives Using Network Analysis». In: *Proceedings of the 4th Joint Workshop on Bibliometric-enhanced Information Retrieval and Natural Language Processing for Digital Libraries co-located with the 42nd International ACM SIGIR Conference on Research and Development in Information Retrieval*. Vol. 2414. CEUR Workshop Proceedings. Paris, France: CEUR-WS.org, lug. 2019, 70–82.
- 34 **Michael Soprano** e Stefano Mizzaro. «Crowdsourcing Peer Review in the Digital Humanities?» In: *Book of Abstracts, 8th AIUCD Conference 2019 – Pedagogy, Teaching, and Research in the Age of Digital Humanities*. AIUCD '19. Udine, Italy, mar. 2019, p. 251. ISBN: 9788894253535.
- 35 **Michael Soprano** e Stefano Mizzaro. «Crowdsourcing Peer Review: As We May Do». In: *Digital Libraries: Supporting Open Science*. Vol. 988. Communications in Computer and Information Science. Pisa, Italy: Springer, gen. 2019, 259–273. DOI: {10.1007/978-3-030-11226-4\_21}.
- 36 Kevin Roitero, **Michael Soprano** e Stefano Mizzaro. «Effectiveness Evaluation with a Subset of Topics: A Practical Approach». In: *The 41st International ACM SIGIR Conference on Research and Development in Information Retrieval*. SIGIR '18. Conference Ranks: GGS A++, CORE A\*. Ann Arbor, MI, USA: Association for Computing Machinery, giu. 2018, pp. 1145–1148. ISBN: 9781450356572. DOI: 10.1145/3209978.3210108.

### Preprints and PhD Thesis

- 37 **Michael Soprano**, Andrea Cioci e Stefano Mizzaro. *Beyond Seeing Is Believing: On Crowdsourced Detection of Audiovisual Deepfakes*. arXiv. Accepted at ROMCIR 2026, held in conjunction with ECIR 2026. Mag. 2026. DOI: 10.48550/arXiv.2605.04797. eprint: 2605.04797 (cs.IR).
- 38 **Michael Soprano**. *In Crowd Veritas: Leveraging Human Intelligence to Fight Misinformation*. arXiv. Revised version of the PhD thesis, defended 22 May 2023. Giu. 2025. DOI: 10.48550/arXiv.2506.09221. eprint: 2506.09221 (cs.IR).

### Bibliometrics

|              |  Scholar |  Scopus |  ACM DL |
|--------------|---------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Publications | 38                                                                                          | 31                                                                                         | 26                                                                                          |
| Citations    | 700                                                                                         | 348                                                                                        | 191                                                                                         |
| h-index      | 14                                                                                          | 9                                                                                          | –                                                                                           |
| i10-index    | 15                                                                                          | –                                                                                          | –                                                                                           |

### Teaching

- 24/25–25/26 **Web Applications.** Role: *Course Instructor*. Master's degree programme in Management Engineering, University of Udine. Course design and teaching; 24 lectures (48 hours). Assessment based on a full-stack project and discussion.
- 24/25–25/26 **Information Analysis and Processing for Training.** Role: *Course Instructor*. Bachelor's degree programme in Sports Science and Master's degree programme in Sciences and Techniques of Preventive and Adapted Physical Activities, University of Udine. Course design and teaching; 12 lectures (24 hours) across the two courses. Assessment based on multiple-choice questions for the Bachelor's course and open-ended questions for the Master's course.
- 20/21–25/26 **Social Computing.** Role: *Teaching Assistant*. Bachelor's degree programme in Internet of Things, Big Data, and Machine Learning, University of Udine. Instructor: Stefano Mizzaro. Preparation and delivery of lectures, slides, lab activities, and assignments; 1–6 lectures (2–12 hours) per academic year.

## Academic Activity

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### Organization

- 2025
- Workshop “AI, Misinformation and Disinformation”, held as part of the 5th National CINI Conference on Artificial Intelligence (Ital-IA 2025), Trieste, June 23. **Organizer.**
  - 21st Italian Research Conference on Digital Libraries (IRCDL 2025), Udine, February 20–21. **Website Chair.**  
<https://ircdl2025.uniud.it/>
- 2024
- 14th Italian Information Retrieval Workshop (IIR 2024), Udine, September 5–6. **Organizer and Website Chair.**  
<https://iir2024.uniud.it/>
- 2023
- CARE: Conversational Agents and Responsible Ethics in Healthcare Workshop, held as part of the European Federation of Medical Informatics Special Topic Conference 2023 (EFMI STC 2023), Turin, October 27. **Organizer.**
- 2022
- Fourth Symposium on Biases in Human Computation and Crowdsourcing (BHCC 2022), On-line, October 12–13. **Proceedings and Website Chair.**

### Program Committee Memberships

- 2025
- 15th Italian Information Retrieval Workshop (IIR 2025), Cagliari, September 3–5. **Full and Short Papers tracks.**
- 2024
- 12th AAAI Conference on Human Computation and Crowdsourcing (HCOMP 2024), Pittsburgh, Pennsylvania, U.S.A. **Full Papers track.**
  - 47th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR 2024), Washington, D.C., U.S.A. **Full and Short Papers tracks.**
- 2023
- 45th European Conference on Information Retrieval (ECIR 2023), Dublin, Ireland. **Short Papers track.**
- 2022
- 45th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR 2022), Madrid, Spain. **Short Papers track.**
- 2021
- Third Symposium on Biases in Human Computation and Crowdsourcing (BHCC 2021), Online. **Program Committee member.**

### Reviewing Activities

- 2026
- 2026 Conference on Empirical Methods in Natural Language Processing (EMNLP 2026), Budapest, Hungary
  - 19th ACM International Conference on Web Search and Data Mining (WSDM 2026), Boise, Idaho, U.S.A.

- 2025
  - *Digital Humanities* (Italian Journal of Digital Humanities), ISSN 2532-8816.  
<https://umanisticadigitale.unibo.it/>
  - *International Journal on Digital Libraries (IJDL)*. <https://portal.issn.org/resource/ISSN/0378-2484>
  - *Data Science and Engineering (DSEJ)*. <https://portal.issn.org/resource/ISSN/23641541>
- 2023
  - *Information Processing & Management (IP&M)*. <https://portal.issn.org/resource/ISSN/0306-4573>. Contribution registered on ORCID profile.
  - *Springer Nature Scientific Reports*. <https://portal.issn.org/resource/ISSN/2045-2322>
  - *13th Italian Information Retrieval Workshop (IIR 2023)*, Pisa, Italy
- 2022
  - *Information Processing & Management*. <https://portal.issn.org/resource/ISSN/0306-4573>
  - *Tenth AAAI Conference on Human Computation and Crowdsourcing (HCOMP 2022)*, Online
  - *Conference and Labs of the Evaluation Forum (CLEF 2022)*, Bologna, Italy
  - *44th European Conference on Information Retrieval (ECIR 2022)*, Stavanger, Norway
  - *Second Symposium on Biases in Human Computation and Crowdsourcing (BHCC 2021)*, Delft, Netherlands (online)
- 2021
  - *30th ACM International Conference on Information and Knowledge Management (CIKM 2021)*, Online
  - *32nd ACM Conference on Hypertext and Social Media (HT 2021)*, Online
- 2020
  - *29th ACM International Conference on Information and Knowledge Management (CIKM 2020)*, Galway, Ireland (online)
  - *31st ACM Conference on Hypertext and Social Media (HT 2020)*, Online
- 2019
  - *15th Italian Research Conference on Digital Libraries (IRCDL 2019)*, Pisa, Italy

## Supervision and Mentoring

- 2020 – Pres. **Supervision:** 1 Bachelor's (BSc) thesis.  
**Co-supervision:** 13 Master's (MSc) and 16 Bachelor's (BSc) theses.  
*Counts include completed and ongoing theses. Additional mentoring activities include advanced student projects, seminars, course assignments, and internships.*

## Affiliations

- 2021 – Pres. **Member** of the Social, Mobile, Data & Crowd Laboratory (SMDC), University of Udine.  
<https://smdc.uniud.it/smdc/en/members/>

## Honors & Awards

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- 2025 **PhD Award 2024.** Award from the University of Udine for the best PhD thesis in the scientific area for 2023. Thesis title: *In Crowd Veritas: Leveraging Human Intelligence to Fight Misinformation*.
- 2022 **Student Grant.** *15th ACM International Conference on Web Search and Data Mining (WSDM 2022)*, Phoenix, U.S.A. Conference registration waiver provided by WSDM. Held remotely due to the COVID-19 pandemic.
- 2020 **Student Grant.** *29th ACM International Conference on Information and Knowledge Management (CIKM 2020)*, Galway, Ireland. Conference registration waiver provided by CIKM. Held remotely due to the COVID-19 pandemic.
- 2020 **Student Volunteer.** *43rd International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR 2020)*, Xi'an, China. Conference registration waiver provided by SIGIR. Held remotely due to the COVID-19 pandemic.

## Links

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**ORCID:** <https://orcid.org/0000-0002-7337-7592>

**GitHub:** <https://github.com/Miccighel>

**Google Scholar:** <https://scholar.google.com/citations?user=ocK0qRUAAAAJ>

**Scopus:** <https://www.scopus.com/authid/detail.uri?authorId=57203392311>

**ACM Digital Library:** <https://dl.acm.org/profile/99659280921>

**Web of Science:** <https://www.webofscience.com/wos/author/record/JDD-6660-2023>

**DBLP:** <https://dblp.org/pid/222/1241.html>

**ResearchGate:** <https://www.researchgate.net/profile/Michael-Soprano>

**arXiv:** [http://arxiv.org/a/soprano\\_m\\_1](http://arxiv.org/a/soprano_m_1)

**Kudos:** [https://www.growkudos.com/profile/michael\\_soprano](https://www.growkudos.com/profile/michael_soprano)

**Semantic Scholar:** <https://www.semanticscholar.org/author/Michael-Soprano/51006308>

**OpenAlex:** <https://openalex.org/authors/A5028918939>

**OSF:** <https://osf.io/fmx3s/>

**Zenodo:** <https://zenodo.org/search?q=michael+soprano>

*I authorize the processing of my personal data in accordance with Regulation (EU) 2016/679 and the Italian Legislative Decree no. 196/2003, as amended by Legislative Decree no. 101/2018, for professional contact purposes.*

Last update: June 19, 2026